

SOLUTION OF THE COUPLED NONLINEAR ALGEBRAIC EQUATIONS APPEARING IN THE FIXED-POINT PERTURBATION THEORY*)

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The old form of the so called fixed-point perturbation theory is reformulated into a formalism suitable for applications. Its structure is described as parallel to the standard Rayleigh-Schrödinger theory. On a number of examples, its algebraic nonlinearity is demonstrated to be almost trivial, i.e. reducible to a finite number of equations (one or two for a few simplest Padé potentials).

1. EQUATIONS

A formal introduction of the model-space projector $P = \sum_{n=1}^N |n\rangle \langle n|$ and of the effective Hamiltonians [1]

$$(1.1) \quad H^{\text{eff}} = H^{\text{eff}}(E, N) = H + (H - E) R(E) (H - E), \\ R(E) = Q(E - QHQ)^{-1} Q, \quad Q = 1 - P$$

is a very useful procedure in the applied quantum theory. Indeed, the partitioned Schrödinger equation

$$(1.2) \quad P(H - E)(P + Q)|\psi\rangle = 0, \\ Q(H - E)(P + Q)|\psi\rangle = 0$$

is precisely equivalent to the $N \times N$ "effective" equation

$$(1.3) \quad H^{\text{eff}}(E, N)|\varphi\rangle = E|\varphi\rangle, \\ |\varphi\rangle = P|\psi\rangle$$

accompanied by the simple definition of wavefunctions

$$(1.4) \quad |\psi\rangle = |\varphi\rangle + R(E)H|\varphi\rangle.$$

Whenever we know H^{eff} , we may solve (1.3) numerically.

The standard variational truncation of H may be characterized as the approximation $R(E) = 0$. Various $R(E) \neq 0$ corrections may be added to PHP by the perturbation techniques. One of them is the so called fixed point perturbation theory (FPPT, [2]), the main idea of which is to construct $R(E)$ as some suitable asymptotic series in the variable $(1/N)^{\text{const}}$. The form of such an ansatz may often be inspired by a geometric "fixed-point" interpretation of transitions between operators H^{eff} for different dimensions N .

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For the sake of definiteness, let us assume that the Hamiltonian is a $2t + 1 -$ diagonal matrix. When partitioned into the $t \times t -$ dimensional submatrices, such a Hamiltonian may be visualised as a block-tridiagonal matrix

$$(1.5) \quad H = \begin{pmatrix} a_0 & b_0 & 0 & \dots & & 0 \\ c_1 & a_1 & b_1 & 0 & \dots & 0 \\ \dots & & & & & \\ 0 & \dots & & 0 & c_k & a_k & b_k & 0 & \dots & 0 \\ \dots & & & & & & & & & \end{pmatrix}.$$

When we put $N = Mt + s$ with $0 < s \leq t$, we may also write (1.1) in a slightly modified notation

$$(1.6) \quad H^{\text{eff}} = \begin{pmatrix} a_0 & b_0 & 0 & \dots & & 0 \\ \dots & & & & & \\ 0 & \dots & & 0 & c_{M-1} & a_{M-1} & b_{M-1} \\ 0 & \dots & & 0 & c_M & a_M & \end{pmatrix}$$

with the $s \times s$ -dimensional submatrix a_0 and $t \times t$ -dimensional effective modification $g_M \neq a_M$ of H . The related identity

$$(1.7) \quad g_M = a_M - b_M \frac{1}{g_{M+1} - E} c_{M+1}$$

gives the well-known "matrix continued fractional" (MCF, [3]) representation of H^{eff} .

The fixed points of the mapping $g_{M+1} \rightarrow g_M$ may be defined by the quadratic equation for matrices $g_M^{[0]} \approx g_M \approx g_{M+1}$,

$$(1.8) \quad g_M^{[0]} = a_M - b_M \frac{1}{g_M^{[0]} - E} c_{M+1}.$$

A solution of the nonlinear equations (1.8) is the main FPPT problem in case of the band-matrix Hamiltonians (1.5) and H^{eff} given via its MCF representants $g_M = g_M^{(0)}$. A subtraction of the approximants (1.8) leads to the iterative scheme with some new MCF quantities $g_M^{(i)}$,

$$(1.9) \quad g_M = g_M^{[0]} + g_M^{(1)} = g_M^{[0]} + g_M^{[1]} + g_M^{(2)} = \dots$$

and with the higher-order fixed-point definitions similar to (1.8) [2].

2. SOLUTIONS

For illustrative purposes, let us consider the radial Schrödinger equation

$$(2.1) \quad \left(-\frac{d^2}{dr^2} + \frac{l(l+1)}{r^2} + V(r) - E \right) \psi(r) = 0$$

with the potential $V(r)$ replaced by its suitable Padé approximations, the simplest forms of which are listed here in table 1.

Table 1.

The first few simplest Padé potentials (a complete list for $t \leq 3$).

$V(r)$	Parameter t	Ref.
$g_0/r = V_C, g_0 r^2 = V_H$	0	[7]
$g/(g_1 + g_2 r) + V_C = V_{C_1}, g/(g_1 + g_2 r^2) + V_H = V_{H_1}$	1	[4]
$g_1 r^{-4/3} + g_2 r^{-2/3} = V_F, g_1 r + V_C, g_1 r^4 + V_H$	2	[6]
$(g + g_1 r)/(g_2 + g_3 r + g_4 r^2) + V_C = V_{C_2}$	2	[8]
$(g + g_1 r^2)/(g_2 + g_3 r^2 + g_4 r^4) + V_H = V_{H_2}$	2	[8]
$g_1 r^{-3/2} + g_2 r^{-1/2} + V_C, g_1 r + g_2 r^2 + V_C, g_1 r^4 + g_2 r^6 + V_H$	3	[9]
$g_3 r + V_{C_1}, g_3 r^4 + V_{H_1}, g/(g_3 + g_4 r^{2/3}) + V_F$	3	—
$(g + g_1 r + g_2 r^2)/(g_3 + g_4 r + g_5 r^2 + g_6 r^3) + V_C = V_{C_3}$	3	—
$(g + g_1 r^2 + g_2 r^4)/(g_3 + g_4 r^2 + g_5 r^4 + g_6 r^6) + V_H = V_{H_3}$	3	—

Up to possible changes of variables of the type $r \rightarrow r^{\text{const}}$, all the Padé-potential forces have a polynomial form

$$(2.2) \quad V_I(r) = \sum_{m=1}^t g_m r^{2m}, \quad g_t > 0$$

or contain the ratio of polynomials. In the latter case, the Hamiltonian itself is a general matrix. Fortunately, in analogy with the elementary V_{H_1} example [4], the Schrödinger equation may still be given an infinite-dimensional band-matrix structure after a modification $H - E \rightarrow \tilde{H} - \tilde{E}$ (= a polynomial $\times (H - E) \times$ a polynomial) in (1.2). In more detail, we shall discuss the simplest nontrivial examples

$$(2.3) \quad (\tilde{H} - \tilde{E})_{II} = (r^2 + U)(p^2 + V_{H_2} - E)(r^2 + V)$$

(cf. table 1, $g_0 = 1$ and $g_2 + g_3 r^2 + g_4 r^4 \sim (r^2 + U)(r^2 + V)$) and

$$(2.4) \quad (\tilde{H} - \tilde{E})_{III}^{(A)} = (r^2 + U)(r^2 + V)(p^2 + V_{H_3} - E)(r^2 + W),$$

$$(\tilde{H} - \tilde{E})_{III}^{(B)} = (r^2 + U)(p^2 + V_{H_3} - E)(r^2 + V)(r^2 + W),$$

in what follows.

Under the above assumptions, we may treat our Schrödinger equations as the difference equations for the projections $\langle n | \psi \rangle$ of wavefunctions on the standard harmonic oscillator states,

$$(2.5) \quad \sum_{n=-t}^t \langle m | (\tilde{H} - \tilde{E}) | m + n \rangle \langle m + n | \psi \rangle = 0, \quad m = 1, 2, \dots$$

In accord with our recent analysis [5], we may write

$$(2.6) \quad \langle n | \psi \rangle = (-1)^n \exp \xi_n$$

where ξ_n is almost constant for large $n \gg 1$. There exist t independent and asymptotically physical solutions of this type – they may be arranged in the matrices, say,

$$(2.7) \quad \tilde{\psi}_{ni} = \langle N - 2t + n | \psi^{(i)} \rangle, \quad \hat{\psi}_{ni} = \langle N - t + n | \psi^{(i)} \rangle, \\ n, i = 1, 2, \dots, t.$$

In this notation, the last t rows of the effective (numerically solvable) equation (1.3) must be satisfied for each column (2.7) separately,

$$(2.8) \quad c_M \tilde{\psi} + g_M \hat{\psi} = 0.$$

Vice versa, each row of these relations may be re-interpreted as a definition of the corresponding row of the effective submatrix g_M ,

$$(2.9) \quad g_M = -c_M \tilde{\psi} (\hat{\psi})^{-1}.$$

This is our basic formula – it remains for us to find the solutions (2.7).

The main idea of the entirely straightforward procedure of solving (2.5) (cf., e.g., the first nontrivial example as described thoroughly in [6]) lies in a use of the ansatz

$$(2.10) \quad \frac{d}{dn} \xi_n = \gamma_1 \varrho + \gamma_2 \varrho^2 + \dots, \quad \varrho = n^{-(1/2)t}.$$

In combination with the Taylor formula

$$(2.11) \quad \exp \xi_{n+k} = \left(1 + k \frac{d}{dn} \xi_n + \dots \right) \exp \xi_n$$

it converts (2.5) into a power-series requirement

$$(2.12) \quad 0 = Z_0(\gamma_1, \gamma_2, \dots) + \varrho Z_1(\gamma_1, \gamma_2, \dots) + \varrho^2 Z_2(\gamma_1, \gamma_2, \dots) + \dots,$$

i.e., into a hierarchy of equations

$$(2.13) \quad Z_k(\gamma_1, \gamma_2, \dots) = 0, \quad k = 0, 1, \dots$$

The structure of their solutions has been analysed via symbolic manipulations on the computer.

In the test run, we have succeeded in reproducing the results obtained in ref. [5] for the polynomial potentials $V_1(r)$ (2.2): For $k = 0, 1, \dots, 2t^2 - 2t - 1$ the identities (2.13) remain trivial and imply that $\gamma_1 = \dots = \gamma_{t-2} = 0$. For $k = 2t(t-1)$, equation (2.13) becomes nonlinear. Its roots specify the $2t$ leading-order coefficients

$$(2.14) \quad \gamma_{t-1} = \left[(-1)^{t+1} \frac{4}{g_t} \right]^{(1/2)t}.$$

Further values $\gamma_t, \gamma_{t+1}, \dots$ follow, in a linear and recurrent way, from the respective $k > 2t(t-1)$ items of (2.13).

Table 2.

Example (2.3) (potential V_{H_2}) and the FPPT recurrences (2.13).

Constraint (2.15)	Index k	Equation $Z_k = 0$	Number of roots	Solutions
(a)	0, 1, ..., 7	linear	—	$\gamma_1 = 0$
	8	nonlinear	four	$\gamma_2^{(a)} = \pm \sqrt{U}$, $\gamma_2^{(b)} = \pm \sqrt{V}$
	9	linear	—	$\gamma_3 = 0$
	10, ...	linear	—	$\gamma_4^{(a)} = -\frac{5}{4}, \dots$, $\gamma_4^{(b)} = -\frac{1}{4}, \dots$
(b)	0, 1, ..., 7	linear	—	$\gamma_1 = 0$
	8	nonlinear	two	$\gamma_2 = \pm \sqrt{U}$
	9, 10, 11	linear	—	$\gamma_3 = 0$
	12	nonlinear	two	$\gamma_4 = -\frac{1}{2} \pm \frac{1}{4}\mu$
	13, ...	linear	—	$\gamma_5 = 0, \dots$
(c)	0, 1, ..., 11	linear	—	$\gamma_1 = \gamma_2 = 0$
	12	nonlinear	four	$\gamma_3 = (-\frac{1}{4}g)^{1/4}$
	13, ...	linear	—	$\gamma_4 = -\frac{3}{8}, \dots$

For the non-polynomial examples (2.3) and (2.4), the results of our analysis are summarized in tables 2 and 3, respectively. In the first case, we were forced to denote $\mu^2 = 1 + g_1 - g/U$ and distinguish between the following three cases

$$(2.15a) \quad 0 \neq U \neq V \neq 0$$

$$(2.15b) \quad 0 \neq U = V, \quad \mu \neq 0$$

$$(2.15c) \quad 0 = U = V, \quad g \neq 0.$$

Again, equations (2.13) are classified as non-linear and linear, and an overall pattern similar to the polynomial case is obtained. Only the case (b) is somewhat exceptional – in place of the complete removal of degeneracy (confluence of the physical and unphysical roots) as encountered before, the $k = 8$ equation (2.13) leaves the results still partially degenerate while the complete separation of all solutions follows from the $k = 12$ equation only.

Qualitatively, the same results are obtained in table 3 and may be expected to take place also in the general case. We may summarize that up to the first few items, the FPPT algebraic equations (2.13) are linear and recurrent definitions of the FPPT

Table 3.
Example (2.4) (potential V_{H_1}) and the FPPT recurrences (2.13).

Constraint	Index k	Equation $Z_k = 0$	Number of roots	Solution (A)	Solution (B)
$0 \neq U \neq V \neq W$	0, 1, ..., 17	linear	—	$\gamma_1 = \gamma_2 = 0$	
	18	nonlinear	six	$\gamma_3^{(a)} = \pm \sqrt{U}, \gamma_3^{(b)} = \pm \sqrt{V}, \gamma_3^{(c)} = \pm \sqrt{W}$	
	19, 20, 21, ...	linear	—	$\gamma_4 = \gamma_5 = 0, \gamma_6^{(a)} = -\frac{5}{4}, \dots$	
$U = V = W \neq 0$ $(\mu = g_2 - g_1[U + g/U^2 \neq 0])$	0, 1, ..., 17	linear	—	$\gamma_1 = \gamma_2 = 0$	
	18	nonlinear	two	$\gamma_3 = \pm \sqrt{U}$	
	19, 20, ..., 23	linear	—	$\gamma_4 = 0$	
	24	nonlinear	three	$\gamma_5 = (\mu\gamma_3/32)^{1/3}$	
	25, ...	linear	—	$\gamma_6^{(A)} = -\frac{7}{12}, \dots, \gamma_6^{(B)} = -\frac{1}{4}, \dots$	
$U = V = W = 0, g \neq 0$	0, 1, ..., 23	linear	—	$\gamma_1 = \gamma_2 = \gamma_3 = 0$	
	24	nonlinear	six	$\gamma_4 = (\frac{1}{4}g)^{1/6}$	
	25, 26, ...	linear	—	$\gamma_5 = 0, \gamma_6^{(A)} = -\frac{1}{2} - \frac{1}{6}(g_1\gamma_4/g), \gamma_6^{(B)} = -\frac{1}{6} - \frac{1}{6}(g_1\gamma_4^3/g)$	

higher-order contributions to the wavefunctions (cf. (2.10)) and energies [via the effective Hamiltonian matrix (1.1) or (1.6) and (2.9)]. This linearity makes them very similar to the standard perturbation expansions [7].

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